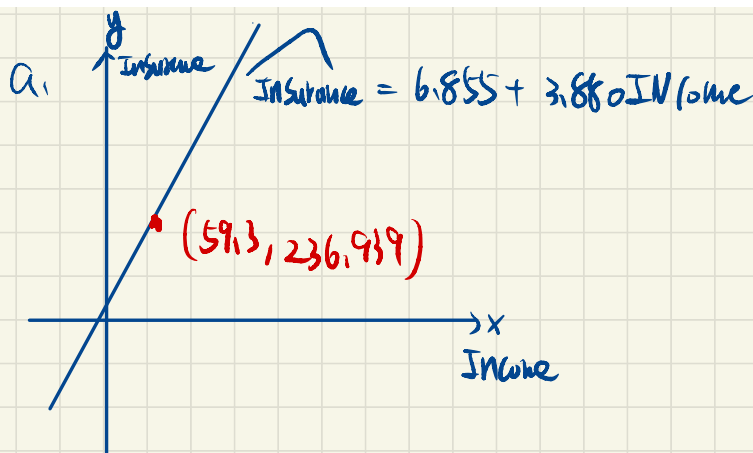


3.18 A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let $INCOME$ be household income (thousands of dollars) and $INSURANCE$ the amount of life insurance held (thousands of dollars). Using a random sample of $N = 20$ households, the least squares estimated relationship is

$$\widehat{INSURANCE} = 6.855 + 3.880 INCOME$$

(se) (7.383) (0.112)

- a. Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of $INCOME = 59.3$. What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.
- b. How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company.



b. When Income + 1000 $\rightarrow$ Insurance + 3.880

$$df = N - 2 = 18 \quad \alpha = 0.05$$

95% CI of β_2 :

$$\hat{\beta}_2 \pm t_{0.025} se(\hat{\beta}_2)$$

$$= 3.880 \pm 2.101 \times 0.112 = [3.645, 4.115]$$

point estimate = 3.880

95% CI = [3.645, 4.115]

- c. Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is -0.746 .
- d. One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be $INSURANCE = \beta_1 + \beta_2 INCOME + e$. Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic used and the rejection region.
- e. Test the hypothesis that as income increases the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion?

$$\widehat{Insurance} = 6.855 + 3.880 \text{ Income} \quad \frac{100000}{1000} = 100$$

$$\begin{aligned}\widehat{Insurance} &= 6.855 + 3.880 \times 100 \\ &= 394.855\end{aligned}$$

$$\begin{aligned}\text{Var}(\widehat{Insurance}) &= \text{Var}(\hat{\beta}_1 + \hat{\beta}_2 x) \\ &= \text{Var}(\hat{\beta}_1) + x^2 \text{Var}(\hat{\beta}_2) + 2x \text{Cov}(\hat{\beta}_1, \hat{\beta}_2) \\ &= 0.383^2 + 100^2 \times 0.112^2 + 2 \times 100 \times (-0.746) \\ &= 30.749\end{aligned}$$

$$\begin{aligned}99\% \text{ CI of } \widehat{Insurance} &= \widehat{Insurance} \pm t_{0.005(18)} \times \text{se}(\widehat{Insurance}) \\ &= 394.855 \pm 2.878 \times \sqrt{30.749} \\ &\approx [378.896, 410.814]\end{aligned}$$

$$d. \begin{cases} H_0: \beta_2 = 5 \\ H_a: \beta_2 \neq 5 \end{cases}$$

$$\alpha = 0.05$$

$$\text{test statistic: } t = \frac{\hat{\beta}_2 - 5}{\text{se}(\hat{\beta}_2)} \sim t(20-2)$$

$$R.R. = \{t, |t| \geq t_{0.025}(18) = 2.101\}$$

$$t_0 = \frac{3.880 - 5}{0.112} = -10$$

$$|t_0| = 10 \geq 2.101, \text{ Reject } H_0$$

We don't have sufficient evidence to conclude that for every \$1000 increase in income, the average amount of life insurance will increase \$5000 at a 95% confidence level.

$$e. \begin{cases} H_0: \beta_2 = 1 \\ H_a: \beta_2 > 1 \end{cases}$$

$$\alpha = 0.01$$

$$\text{test statistic: } t = \frac{\hat{\beta}_2 - 1}{\text{se}(\hat{\beta}_2)} \sim t(20-2)$$

$$R.R. = \{t, t \geq t_{0.01}(18) = 2.552\}$$

$$t = \frac{3.880 - 1}{0.112} \approx 25.714$$

$$t_0 = 25.714 \in R.R., \text{ reject } H_0$$

We have sufficient evidence to conclude that the slope greater than one.

HW0317_314707031

James

2026-03-22

```
library(POE5Rdata)
data("collegetown")
data("tuna")
```

Problem 3.23 - Collegetown House Prices

model: $PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$

```
str(collegetown)
```

```
## 'data.frame': 500 obs. of 8 variables:
## $ price : num 96.5 106 114 119.9 122.7 ...
## $ sqft : num 16.8 17.5 18.4 29.1 14.7 ...
## $ age : int 10 10 9 9 9 8 10 9 7 10 ...
## $ pool : int 0 0 0 0 0 0 0 0 0 0 ...
## $ fireplace: int 0 0 0 0 0 0 0 0 0 0 ...
## $ close : int 0 0 0 0 0 0 0 0 0 0 ...
## $ twostory : int 0 0 0 0 0 0 0 0 0 0 ...
## $ occupied : int 0 0 1 0 1 0 1 1 0 1 ...
```

```
reg_quad <- lm(price ~ I(sqft^2), data = collegetown)
summary(reg_quad)
```

```
##
## Call:
## lm(formula = price ~ I(sqft^2), data = collegetown)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -383.67  -48.39   -7.50   38.75  469.70
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 93.565854   6.072226  15.41  <2e-16 ***
## I(sqft^2)    0.184519   0.005256  35.11  <2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 92.08 on 498 degrees of freedom
## Multiple R-squared:  0.7122, Adjusted R-squared:  0.7117
## F-statistic: 1233 on 1 and 498 DF,  p-value: < 2.2e-16

a1 <- coef(reg_quad)[1]
a2 <- coef(reg_quad)[2]
```

(a) Hypothesis test: marginal effect at SQFT = 20

The marginal effect of SQFT on PRICE is:

$$\frac{dPRICE}{dSQFT} = 2\alpha_2 \cdot SQFT$$

At SQFT = 20 (i.e., 2000 sq ft), increasing by 1 unit of SQFT (100 sq ft):

$$ME = 2\alpha_2 \times 20$$

- $H_0: 2\alpha_2 \times 20 \leq 13$ (i.e., $\alpha_2 \leq 0.325$)
- $H_1: 2\alpha_2 \times 20 > 13$ (i.e., $\alpha_2 > 0.325$)

This is a one-sided (right-tail) t-test.

```
# Marginal effect = dE(PRICE/SQFT)/dSQFT = 2 * a2 * SQFT
# Marginal effect at SQFT = 20: ME = 2 * a2 * 20 = 40 * a2
ME_2000 <- 40 * a2
cat("Estimated marginal effect at SQFT=20:", ME_2000, "(in $1000)\n")
```

```
## Estimated marginal effect at SQFT=20: 7.38076 (in $1000)
```

```
# Test H0: 40*a2 <= 13, equivalently H0: a2 <= 0.325
# t = (a2_hat - 0.325) / se(a2_hat)
se_a2 <- summary(reg_quad)$coefficients[2, 2]
t_stat_a <- (a2 - 0.325) / se_a2
df_a <- reg_quad$df.residual
p_value_a <- 1 - pt(t_stat_a, df = df_a) # right-tail

cat("t-statistic:", t_stat_a, "\n")
```

```
## t-statistic: -26.72875
```

```
cat("Critical value (5%, right-tail):", qt(0.95, df = df_a), "\n")
```

```
## Critical value (5%, right-tail): 1.647919
```

```
cat("p-value:", p_value_a, "\n")
```

```
## p-value: 1
```

```

if (p_value_a < 0.05) {
  cat("Conclusion: Reject H0 at 5% level. The marginal effect is significantly greater than $13,000.\n")
} else {
  cat("Conclusion: Do not reject H0 at 5% level. We don't have sufficient evidence to conclude that t
}

```

```
## Conclusion: Do not reject H0 at 5% level. We don't have sufficient evidence to conclude that the marg
```

(b) Hypothesis test: marginal effect at SQFT = 40

At SQFT = 40 (i.e., 4000 sq ft), increasing by 1 unit of SQFT (100 sq ft):

$$ME = 2\alpha_2 \times 40 = 80\alpha_2$$

- $H_0: 80\alpha_2 \leq 13$ (i.e., $\alpha_2 \leq 0.1625$)
- $H_1: 80\alpha_2 > 13$ (i.e., $\alpha_2 > 0.1625$)

```

ME_4000 <- 80 * a2
cat("Estimated marginal effect at SQFT=40:", ME_4000, "(in $1000)\n")

```

```
## Estimated marginal effect at SQFT=40: 14.76152 (in $1000)
```

```

# Test H0: a2 <= 0.1625
t_stat_b <- (a2 - 0.1625) / se_a2
p_value_b <- 1 - pt(t_stat_b, df = df_a)

cat("t-statistic:", t_stat_b, "\n")

```

```
## t-statistic: 4.189467
```

```
cat("Critical value (5%, right-tail):", qt(0.95, df = df_a), "\n")
```

```
## Critical value (5%, right-tail): 1.647919
```

```
cat("p-value:", p_value_b, "\n")
```

```
## p-value: 1.65395e-05
```

```

if (p_value_b < 0.05) {
  cat("Conclusion: Reject H0 at 5% level. The marginal effect is significantly greater than $13,000.\n")
} else {
  cat("Conclusion: Fail to reject H0 at 5% level. e don't have sufficient evidence to conclude that the
}

```

```
## Conclusion: Reject H0 at 5% level. The marginal effect is significantly greater than $13,000.
```

(c) 95% interval estimate of $E(\text{PRICE}|\text{SQFT}=20)$

$$E(\text{PRICE}|\text{SQFT} = 20) = \alpha_1 + \alpha_2 \times 20^2 = \alpha_1 + 400\alpha_2$$

```
new_data <- data.frame(sqft = 20)
pred_c <- predict(reg_quad, newdata = new_data, interval = "confidence", level = 0.95)
pred_c
```

```
##          fit          lwr          upr
## 1 167.3735 158.0481 176.6988
```

Interpretation of Estimates:

1. Expected Price & Confidence Interval: Based on the quadratic regression model $E(PRICE|SQFT) = \alpha_1 + \alpha_2 SQFT^2$, the expected selling price for a 2,000 square foot house ($sqft = 20$) is estimated to be 167.3735 (in thousands of dollars). The 95% confidence interval for this expected price is [158.0481, 176.6988].
2. Description for a General Audience: "In repeated sampling, if we were to draw many samples from the population and construct a confidence interval in the same way, 95% of such intervals would contain the true average market selling price for a 2,000 square foot house. Based on our specific sample, this interval is calculated to be between \$158,048 and \$176,699."

(d) Sample mean of houses with SQFT = 20

```
houses_2000 <- collegetown[collegetown$sqft == 20, ]
cat("Number of houses with SQFT = 20:", nrow(houses_2000), "\n")
```

```
## Number of houses with SQFT = 20: 3
```

```
if (nrow(houses_2000) > 0) {
  sample_mean <- mean(houses_2000$price)
  cat("Sample mean of selling price:", sample_mean, "(in $1000)\n\n")

  cat("Comparison with part (c):\n")
  cat(" Predicted E(PRICE|SQFT=20):", pred_c[1], "\n")
  cat(" Sample mean:", sample_mean, "\n")
  cat(" 95% CI: [", pred_c[2], ",", pred_c[3], "]\n\n")
}
```

```
## Sample mean of selling price: 163.3333 (in $1000)
##
## Comparison with part (c):
## Predicted E(PRICE|SQFT=20): 167.3735
## Sample mean: 163.3333
## 95% CI: [ 158.0481 , 176.6988 ]
```

Interpretation of Estimates:

1. Sample Calculation In our dataset, there are exactly 3 houses with a living area of 2,000 square feet ($sqft = 20$). The actual sample mean of the selling prices for these 3 houses is 163.3333 (in thousands of dollars).

2. Compatibility & Explanation This calculated sample mean (163.3333) falls well within the 95% confidence interval [158.0481, 176.6988] that we estimated in part (c).

This indicates that the actual observed market data is highly compatible with our model's prediction. The quadratic regression model successfully and reasonably captures the expected price level for houses of this specific size.

Problem 3.31 - Tuna Sales

```
str(tuna)
```

```
## 'data.frame': 52 obs. of 6 variables:
## $ sal1 : int 6439 3329 3415 2909 2598 3773 20383 11761 2614 2496 ...
## $ apr1 : num 0.66 0.62 0.62 0.62 0.63 0.69 0.63 0.65 0.77 0.81 ...
## $ apr2 : num 0.82 0.8 0.77 0.66 0.65 0.63 0.65 0.78 0.62 0.69 ...
## $ apr3 : num 0.79 0.59 0.63 0.81 0.81 0.8 0.78 0.8 0.77 0.68 ...
## $ disp : int 1 1 0 0 0 0 0 1 1 1 ...
## $ displ: int 0 0 0 0 0 0 1 0 0 0 ...
```

(a) Summary statistics for SAL1 and APR1

```
library(knitr)

summary_table <- data.frame(
  Variable = c("SAL1 (Unit Sales)", "APR1 (Price per can, $)"),
  Mean = c(mean(tuna$sal1), mean(tuna$apr1)),
  Min = c(min(tuna$sal1), min(tuna$apr1)),
  Max = c(max(tuna$sal1), max(tuna$apr1)),
  Std_Dev = c(sd(tuna$sal1), sd(tuna$apr1))
)

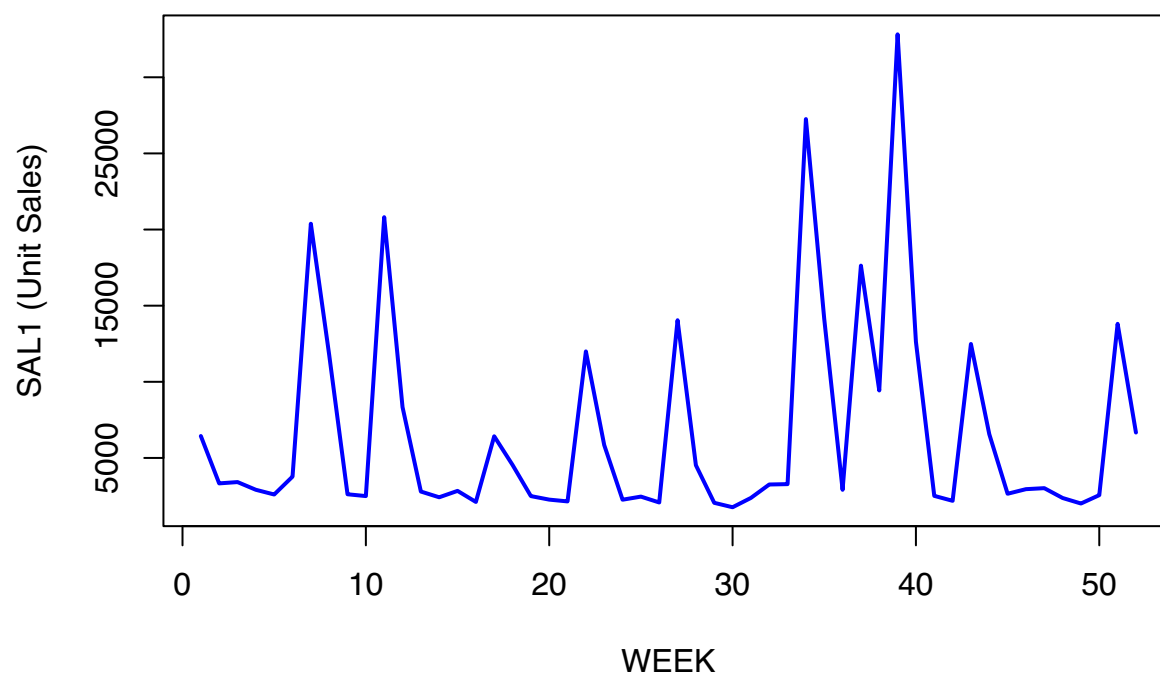
kable(summary_table, digits = 2, caption = "Summary Statistics for SAL1 and APR1")
```

Table 1: Summary Statistics for SAL1 and APR1

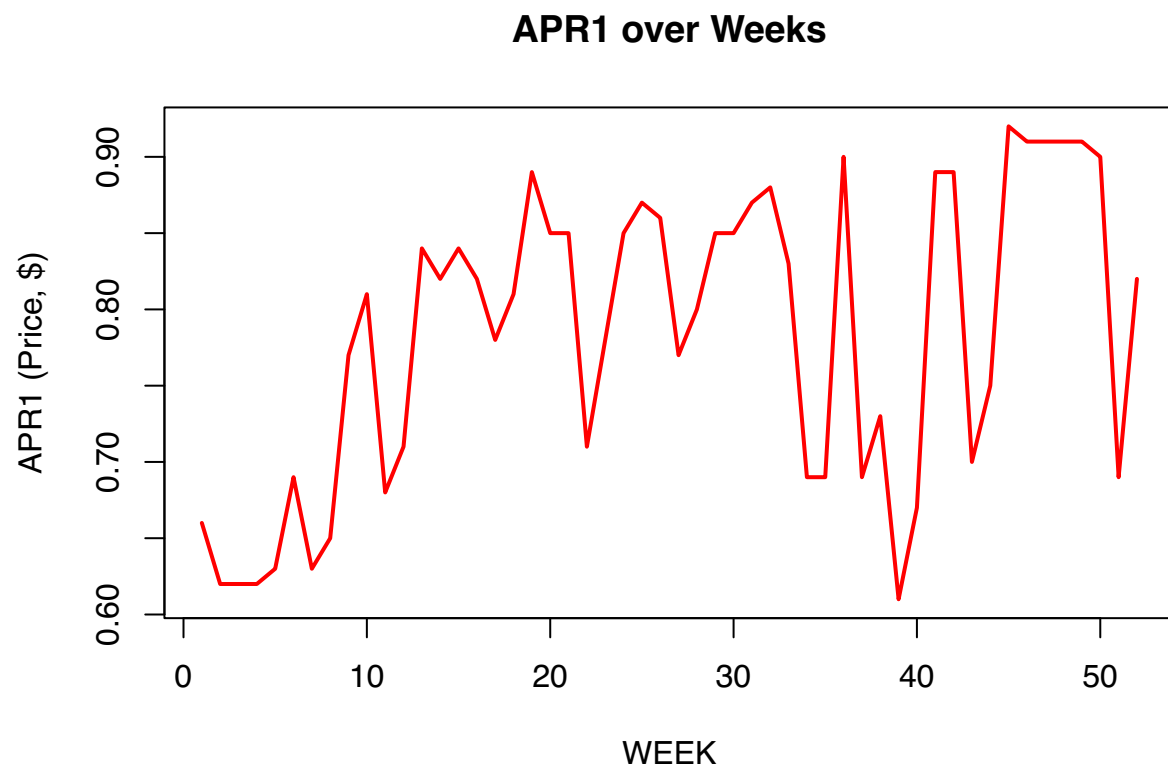
Variable	Mean	Min	Max	Std_Dev
SAL1 (Unit Sales)	6718.71	1762.00	32820.00	6915.08
APR1 (Price per can, \$)	0.78	0.61	0.92	0.10

```
# Plot SAL1 vs WEEK
plot(tuna$sal1, type = "l", col = "blue", lwd = 2,
     main = "SAL1 over Weeks", xlab = "WEEK", ylab = "SAL1 (Unit Sales)")
```


SAL1 over Weeks

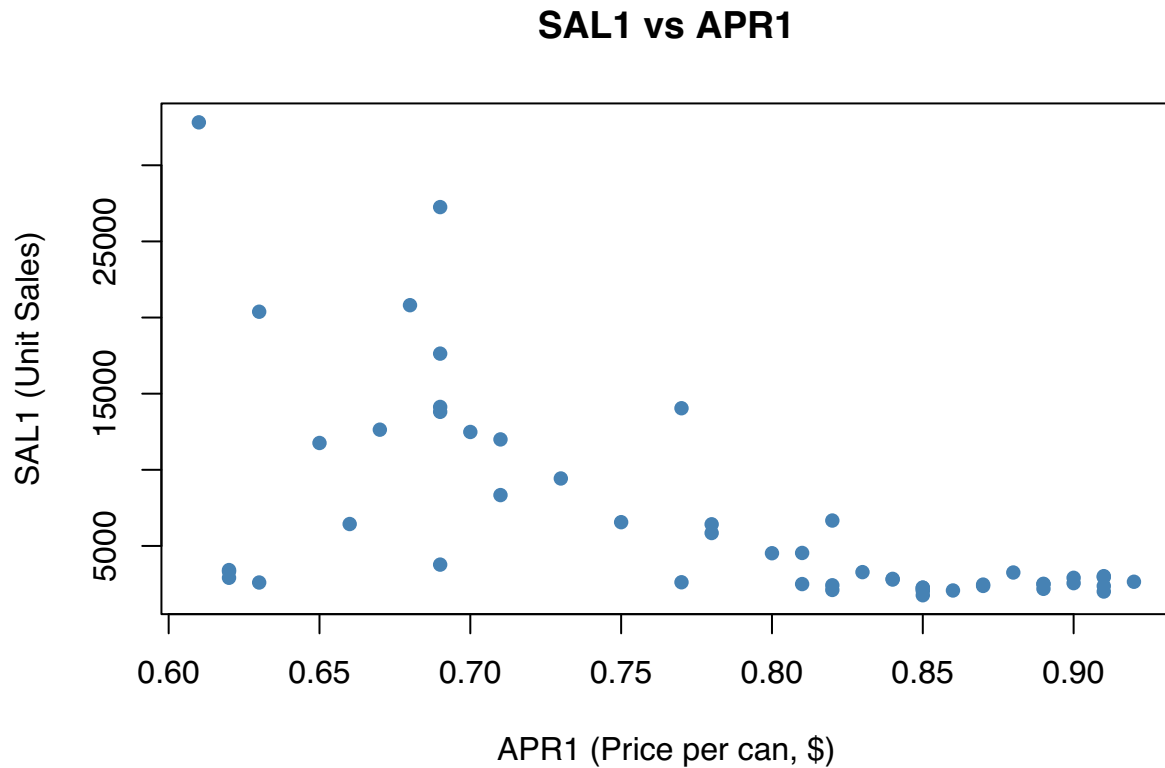


```
# Plot APR1 vs WEEK
plot(tuna$apr1, type = "l", col = "red", lwd = 2,
     main = "APR1 over Weeks", xlab = "WEEK", ylab = "APR1 (Price, $)")
```



(b) Plot SAL1 (y) vs APR1 (x)

```
plot(tuna$apr1, tuna$sal1,  
     pch = 16, col = "steelblue",  
     main = "SAL1 vs APR1",  
     xlab = "APR1 (Price per can, $)",  
     ylab = "SAL1 (Unit Sales)")
```



Interpretation of Estimates:

1. The Relationship There is a clear negative relationship between SAL1 (Unit Sales) and APR1 (Price per can). We can see that when the price is lower (around \$0.60 to \$0.70), the unit sales are much higher and more volatile. As the price increases (towards \$0.80 and \$0.90), the sales volume drops off significantly and flattens out near the bottom of the y-axis.
2. Is this expected? Yes, this is exactly what we would expect.
3. Why? This perfectly illustrates the fundamental economic Law of Demand. When the price of a normal good (like a can of tuna) goes up, consumers typically buy less of it. This happens for two main reasons:

(c) **Linear regression:** $SAL1 = \beta_1 + \beta_2 \cdot PRICE1 + e$

where $PRICE1 = 100 \times APR1$.

```
tuna$price1 <- 100 * tuna$apr1
reg_tuna <- lm(sal1 ~ price1, data = tuna)
summary(reg_tuna)
```

```
##
## Call:
## lm(formula = sal1 ~ price1, data = tuna)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10869.5  -1588.3   -499.3   1626.2  18607.1
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 40714.22    6196.42   6.571 2.82e-08 ***
## price1      -434.45     78.58  -5.528 1.17e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5502 on 50 degrees of freedom
## Multiple R-squared:  0.3794, Adjusted R-squared:  0.367
## F-statistic: 30.56 on 1 and 50 DF,  p-value: 1.172e-06
```

```
b1 <- coef(reg_tuna)[1]
b2 <- coef(reg_tuna)[2]
se_b2 <- summary(reg_tuna)$coefficients[2, 2]
df_tuna <- reg_tuna$df.residual
```

```
confint(reg_tuna, "price1", level = 0.95)
```

```
##           2.5 %    97.5 %
## price1 -592.2897 -276.6049
```

Interpretation of Estimates:

1. Estimated Regression Equation: Based on the regression output, the estimated linear equation is: $\widehat{SAL1} = 40714.22 - 434.45 \times PRICE1$
2. Interpretation of the Point Estimate ($\hat{\beta}_2$): The estimated coefficient for $PRICE1$ is -434.45 . This indicates that, on average, a one-cent increase in the price of a can of tuna is associated with a decrease of approximately 434.45 units in weekly sales.
3. Interpretation of the 95% Confidence Interval: The calculated 95% confidence interval for β_2 is $[-592.29, -276.60]$. In repeated sampling, if we were to draw many samples from the population and construct confidence intervals in the same way, 95% of such intervals would contain the true marginal effect of a one-cent price increase. Based on our current sample, this calculated interval suggests that a one-cent increase in price reduces weekly sales by an amount between roughly 276.60 and 592.29 units.

(d) 90% interval estimate for $E(SAL1|PRICE1 = 70)$

When price per can is 70 cents:

```
new_tuna <- data.frame(price1 = 70)
pred_d <- predict(reg_tuna, newdata = new_tuna, interval = "confidence", level = 0.90)
pred_d
```

```
##      fit      lwr      upr
## 1 10302.9 8624.934 11980.87
```

```
cat("Expected sales at price = 70 cents:", pred_d[1], "\n")
```

```
## Expected sales at price = 70 cents: 10302.9
```

```
cat("90% Confidence Interval: [", pred_d[2], ",", pred_d[3], "]\n")
```

```
## 90% Confidence Interval: [ 8624.934 , 11980.87 ]
```

(e) 95% CI for elasticity at the means

$$\varepsilon = \beta_2 \times \frac{\overline{PRICE1}}{\overline{SAL1}}$$

Treating the sample means as constants:

```
mean_price1 <- mean(tuna$price1)
mean_sal1 <- mean(tuna$sal1)

elasticity <- b2 * (mean_price1 / mean_sal1)

se_elasticity <- (mean_price1 / mean_sal1) * se_b2

t_crit_95 <- qt(0.975, df = df_tuna)
ci_elast_lower <- elasticity - t_crit_95 * se_elasticity
ci_elast_upper <- elasticity + t_crit_95 * se_elasticity

cat("Elasticity at means:", elasticity, "\n")
```

```
## Elasticity at means: -5.059825
```

```
cat("95% CI for elasticity: [", ci_elast_lower, ",", ci_elast_upper, "]\n")
```

```
## 95% CI for elasticity: [ -6.898149 , -3.221501 ]
```

Interpretation of Estimates:

1. Point Estimate of Elasticity The estimated price elasticity of demand for brand no. 1 tuna, evaluated at the sample means ($\overline{PRICE1} = 78.25$, $\overline{SAL1} = 6718.71$), is approximately -5.06.
2. 95% Interval Estimate Treating the sample means as constants, the calculated 95% confidence interval for this elasticity is [-6.898, -3.222]. In repeated sampling, 95% of such constructed intervals would contain the true price elasticity at the mean. Our calculated interval suggests we are 95% confident that the true elasticity lies within this range.
3. Economic Interpretation: Elastic or Inelastic? Since the absolute value of the estimated elasticity is significantly greater than 1 ($|-5.06| > 1$), and the entire 95% confidence interval lies well below -1, we can conclude that the demand for brand no. 1 tuna is highly elastic with respect to price. This means a 1% increase in its price is expected to lead to a decrease in sales of approximately 5.06%.

4. Does this make economic sense? Yes, this makes perfect economic sense. Brand no. 1 tuna is a specific brand of a common grocery item (canned tuna). In a typical supermarket, there are many close substitutes available (e.g., brand no. 2, brand no. 3, or other types of canned fish/meat). When consumers have many easy substitute options, they become highly sensitive to price changes for any single brand. If brand no. 1 raises its price even slightly, consumers can easily switch to a cheaper competing brand, leading to a drastic drop in sales for brand no. 1.

(f) Test hypothesis: elasticity = -3

- $H_0: \varepsilon = -3$, i.e., $\beta_2 \times \frac{\overline{PRICE1}}{\overline{SAL1}} = -3$, i.e., $\beta_2 = -3 \times \frac{\overline{SAL1}}{\overline{PRICE1}}$
- $H_1: \varepsilon \neq -3$

Using 10% significance level (two-tailed test).

```
# H0: elasticity = -3, equivalently b2 = -3 * mean_sal1 / mean_price1
b2_null <- -3 * mean_sal1 / mean_price1
cat("Under H0, b2 =", b2_null, "\n")
```

```
## Under H0, b2 = -257.5864
```

```
t_stat_f <- (b2 - b2_null) / se_b2
p_value_f <- 2 * pt(-abs(t_stat_f), df = df_tuna) # two-tailed
t_crit_10 <- qt(0.95, df = df_tuna)

cat("t-statistic:", t_stat_f, "\n")
```

```
## t-statistic: -2.250572
```

```
cat("Critical values (10%, two-tailed): +/-", t_crit_10, "\n")
```

```
## Critical values (10%, two-tailed): +/- 1.675905
```

```
cat("Rejection region: |t| >", t_crit_10, "\n")
```

```
## Rejection region: |t| > 1.675905
```

```
cat("p-value:", p_value_f, "\n\n")
```

```
## p-value: 0.02884204
```

Interpretation of Estimates:

Conclusion: The calculated p-value is less than the 0.10 significance level ($p < 0.10$). Therefore, we reject the null hypothesis. We have sufficient statistical evidence at the 10% level to conclude that the price elasticity of demand is significantly different from -3.